CSE4203 Principles of Software Architecture

RapidMed

Emergency Triage and Real-Time Ambulance Dispatch System

Milestone 1: Proposal and architectural requirements

Team	RapidMed
Course	CSE4203 Principles of Software Architecture
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Team members and contributions

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1. System overview

The problem it solves

An ambulance crew with a seriously ill or injured patient needs to know which emergency departments can take the patient now and how long each will take to reach, and the receiving team needs to know what is coming. Without a shared, live view of capacity, the choice rests on distance and phone calls.

RapidMed gives the crew, the hospital and the dispatch centre the same live picture. On the ambulance's computer, the paramedic records vital signs and a triage level in an app that works without a signal, and opens the patient's emergency summary from Sanad, the patient health data management system (PHDMS). RapidMed ranks nearby hospitals by free beds of the type needed and by traffic-aware travel time, reserves a bed at the chosen one and sends the patient's details ahead, while a dispatch supervisor watches the fleet and can override any routing decision.

Who uses it

User and the part they use	What they do
Field paramedic Ambulance computer app	Records vitals and triage level, even offline; opens the patient's summary (break-glass if needed); gets the destination, route and alerts
Hospital triage coordinator Web portal, hospital view	Keeps bed availability up to date; accepts incoming patients and prepares the team using their vital signs and arrival time
Central dispatch supervisor Web portal, dispatch view	Monitors ambulances and hospital capacity on a live map; overrides routing; acts on alerts, such as a crew losing its connection

Why it is architecturally interesting

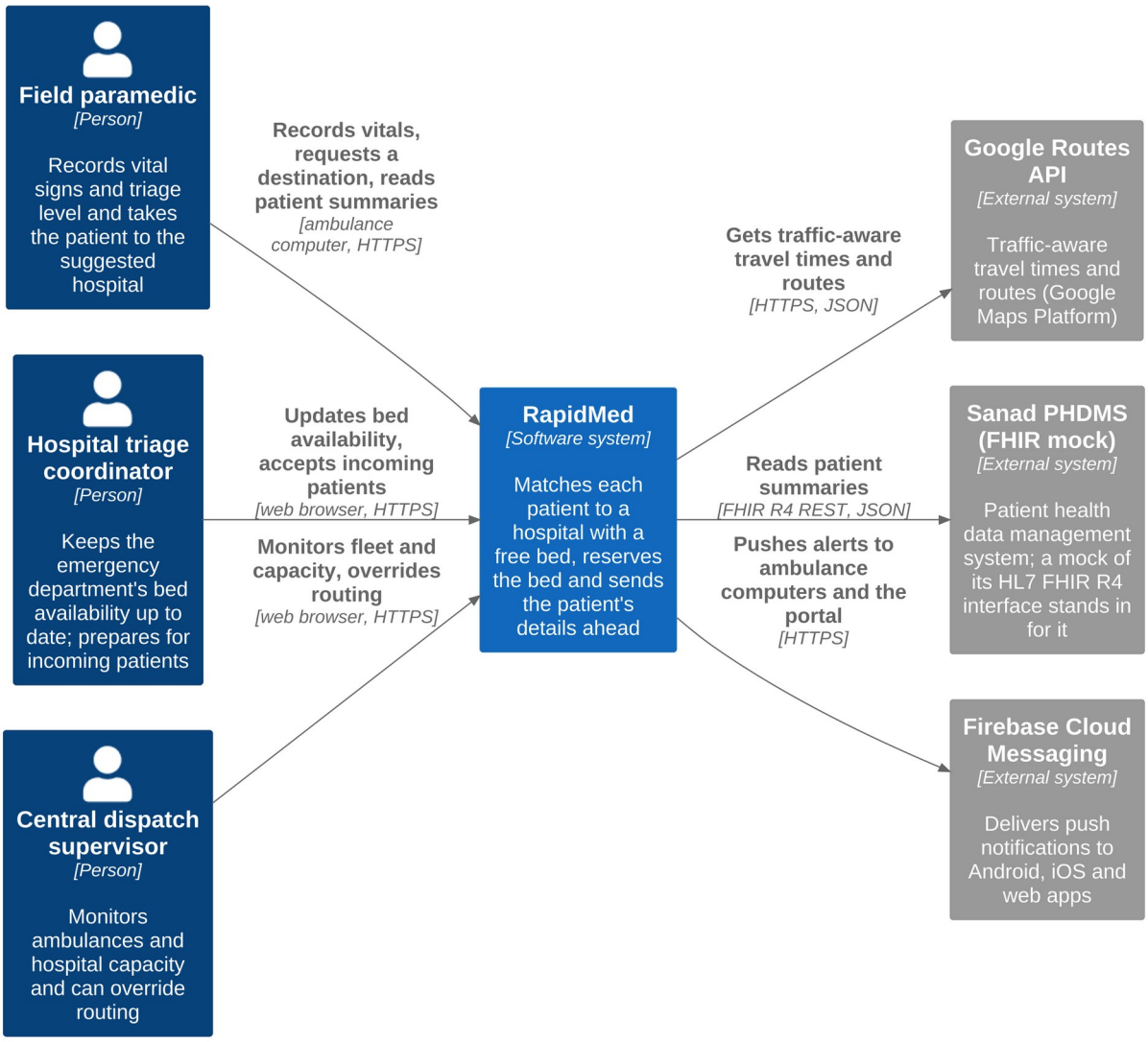
1. **Consistency against availability.** A bed can go to only one patient, so reservations must be strictly consistent: two crews must never be sent to the last free resuscitation bay. Yet crews work on roads with weak or no signal and must keep recording and reading data offline. When the network is partitioned, a system cannot guarantee both, so the design must decide which actions may go ahead offline and which must wait for the server.
2. **Security and privacy against usability and speed.** Health data are sensitive personal data under Egypt's data protection law, and patient histories need encryption, role-based access and an audit trail. But an unconscious patient cannot consent, and the crew cannot wait for approval: the record must open within seconds through a "break-glass" path that overrides normal access control while recording who used it and why. Caching records on the ambulance's computer helps speed and offline work but leaves sensitive data on a device shared by several crews.
3. **Dependence on outside services.** Routing, patient summaries and notifications rely on three external services, any of which can slow down, fail or hit a usage limit. None of these failures may stop a crew from getting a destination, so the design needs time-outs, fallbacks and circuit breakers, and a mock of each service to test failures.

2. Stakeholders and concerns

Stakeholder	Key concerns
Field paramedics End users: app on the ambulance's computer	• The app keeps working with a weak or no signal and never loses entered data • Recording vital signs takes a few taps, even with gloves on • A destination within seconds of asking • The patient's allergies, conditions and medications on screen quickly, even when the patient cannot speak
Hospital triage coordinators End users: web portal	• Two critical patients are never sent for one remaining bed • Crews see the capacity the coordinator set, and how old it is • Early notice of each incoming patient, with vital signs and arrival time • Updating capacity takes seconds during a busy shift
Central dispatch supervisors End users: web portal	• A live, accurate view of every ambulance and hospital • Overrides reach the crew at once and are confirmed • Alerts when a crew loses its connection or a hospital fills • The reason behind each recommendation can be seen
Patients and their families Data subjects, not users	• Taken without delay to a hospital that can treat them • Medical history seen only by those treating them • Every emergency access recorded and open to review
Ambulance service and hospital management Business owners	• Fewer wasted trips and shorter handovers • Reports on transport times, reservations and overrides • Little extra work for crews and hospital staff • Low running costs for hosting and API calls
Personal Data Protection Center Regulator, Egypt	• Health data are sensitive personal data under Law No. 151 of 2020 • Access only by those who need it, for a stated purpose • Data protected in storage and in transit • Breaches reported to the Center within 72 hours • Break-glass use justified, logged and reviewed
Sanad PHDMS owners Owners of the patient health data management system that RapidMed reads from	• Only registered systems and authorised staff can query records • Every query traceable to a person and a reason • Standard FHIR requests, within agreed rate limits
Operations staff Operators	• Each of the three parts deployed and rolled back on its own • Monitoring and alerts for each part and each external service • API keys and other secrets kept out of the code • Graceful behaviour when an external service fails
Development team Developers (this team)	• Clear boundaries, so that three people can work in parallel • Every external service replaceable by a mock for testing • Free service tiers enough for the prototype • Milestone deadlines met

3. System context

System context diagram for RapidMed (C4 Level 1)



Legend

Person	A user role that works with RapidMed
Software system	The system in scope, shown as a single box
External system	A software system outside RapidMed's boundary
→	Relationship, drawn from the element that starts the interaction; label: purpose [technology]

4. Functional requirements

ID	MoSCoW priority	User story	In M3 prototype
US-01	Must	As a **field paramedic**, I want to sign in to the shared ambulance computer with my own account, so that what I record and open is attributed to me and kept apart from other crews' data.	Yes
US-02	Must	As a **field paramedic**, I want to record vital signs and a triage level in a few taps, with or without a signal, so that data entry never takes my attention from the patient or gets lost.	Yes
US-03	Must	As a **field paramedic**, I want a recommended hospital with a reserved bed within seconds of asking, so that I can leave for the right hospital without phone calls.	Yes
US-04	Must	As a **field paramedic**, I want the patient's allergies, conditions and medications on screen quickly, through break-glass if the patient cannot consent, so that I avoid giving a harmful treatment.	Yes
US-05	Must	As a **hospital triage coordinator**, I want two critical patients never to be sent for one remaining bed, so that every patient we accept has a bed waiting.	Yes
US-06	Must	As a **hospital triage coordinator**, I want crews to see the capacity I entered and how recent it is, so that they decide on information they can judge.	No
US-07	Must	As a **hospital triage coordinator**, I want early notice of each incoming patient with vital signs and arrival time, so that my team is ready when the ambulance arrives.	Yes
US-08	Should	As a **hospital triage coordinator**, I want to update capacity in seconds during a busy shift, so that the figures stay accurate without slowing my work.	No
US-09	Should	As a **hospital triage coordinator**, I want a reserved bed released automatically if the ambulance neither arrives nor renews the reservation in time, so that beds are not held for patients who went elsewhere.	No
US-10	Must	As a **central dispatch supervisor**, I want a live view of every ambulance and hospital on one map, so that I can see the whole situation in one place.	No
US-11	Must	As a **central dispatch supervisor**, I want my routing overrides to reach the crew at once and be confirmed, so that I know the crew has acted on the change.	No
US-12	Should	As a **central dispatch supervisor**, I want alerts when a crew loses its connection or a hospital fills, so that I can step in before it affects a patient.	No
US-13	Should	As a **central dispatch supervisor**, I want to review every break-glass access, so that emergency access stays justified and questions from the regulator can be answered.	No
US-14	Could	As a **central dispatch supervisor**, I want to see why a hospital was recommended, so that I can decide whether to override it.	No
US-15	Won't (this time)	As a **hospital triage coordinator**, I want bed counts read automatically from the hospital information system, so that I do not have to enter them.	No

5. Quality attribute scenarios

QAS-1. Availability: no data lost offline

Part	Description
Source	Field paramedic
Stimulus	Records vital signs and a triage level while the ambulance computer has no network connection
Artifact	Paramedic app and its local store
Environment	No connection for up to 30 minutes; the app is closed and reopened during the outage
Response	Each entry is saved on the device before the app confirms it; entries are queued and sent in order when the connection returns; the server ignores any entry it has already received
Response measure	In 20 test outages of 1 to 30 minutes, with 500 entries in total and 5 forced app closures: no entry lost, no entry duplicated on the server, and every queued entry stored on the server within 30 s of the connection returning

QAS-2. Availability: routing service failure

Part	Description
Source	Google Routes API (external system)
Stimulus	Stops answering: every request times out or returns an error
Artifact	Routing adapter of the dispatch and routing engine
Environment	Normal operation, 20 destination requests per minute
Response	Calls time out after 1.5 s; after 3 consecutive failures a circuit breaker opens, and the engine ranks hospitals using travel times cached in the last 30 minutes, or straight-line distance where none is cached, and labels the result “estimated”; it tries the service again every 30 s and closes the breaker when a call succeeds
Response measure	During a 10-minute simulated outage, 100% of destination requests return a ranked result labelled “estimated” within 3 s; once the breaker is open, no request waits for the service; normal ranking returns within 60 s of the service recovering

QAS-3. Performance: destination at peak load

Part	Description
Source	100 simulated ambulance crews
Stimulus	Request a destination: 200 requests within 60 s
Artifact	Dispatch and routing engine
Environment	Peak load; 25 candidate hospitals; the Routes API is replaced by a mock that answers in 300 ms
Response	Ranks the hospitals by free beds of the type needed and by travel time, and returns the top three with an offer to reserve
Response measure	95% of requests answered within 3 s and 99% within 5 s, measured at the app, with no errors

QAS-4. Consistency (safety): the last bed

Part	Description
Source	Two field paramedics
Stimulus	Both request the last free resuscitation bed at the same hospital within 100 ms; one request is retried after a network time-out
Artifact	Reservation component of the dispatch and routing engine, and its database
Environment	Normal operation
Response	Exactly one reservation succeeds; the retry returns the same result instead of creating a second reservation; the other crew is told at once and offered the next-best hospital
Response measure	Over 1,000 runs, 0 double reservations and 0 extra reservations from retries; the losing crew sees an alternative within 3 s in 95% of runs

QAS-5. Security: break-glass access

Part	Description
Source	An authenticated paramedic treating an unconscious patient
Stimulus	Opens the patient’s summary through break-glass, giving a reason
Artifact	Break-glass path and audit log of the dispatch and routing engine
Environment	Normal operation; the patient cannot consent
Response	The summary opens; an audit record of the user, ambulance, patient, reason and time is written to an append-only log and flagged for review
Response measure	The summary opens within 3 s in 95% of 200 test requests; 100% of them have a complete audit record within 5 s; 0 of 50 requests without a reason succeed; 0 of 20 attempts to edit or delete an audit record succeed from any application role; one month’s break-glass accesses are listed in under 1 minute

QAS-6. Interoperability: Sanad under a rate limit

Part	Description
Source	15 ambulance crews
Stimulus	90 patient summary requests within 60 s
Artifact	Sanad FHIR client of the dispatch and routing engine
Environment	Peak load; Sanad’s limit is assumed to be 60 requests per minute
Response	Sends standard HL7 FHIR R4 requests carrying RapidMed’s registered client credentials, the user and the reason; sends at most 60 per minute and queues the rest; answers from the episode cache when a summary was already fetched
Response measure	In a 10-minute test, 0 requests above 60 per minute and 0 rate-limit errors from the Sanad mock; 100% of 200 sampled requests are valid FHIR R4 and carry user and reason; 95% of the first 60 requests are answered within 3 s and the other 30 within 60 s; the mock rejects all 50 requests sent without valid client credentials

QAS-7. Usability: recording vital signs

Part	Description
Source	A field paramedic who has not used RapidMed before
Stimulus	Records heart rate, blood pressure, respiratory rate, oxygen saturation and a triage level
Artifact	Vital-signs screen of the paramedic app
Environment	Normal operation; the paramedic wears examination gloves; the computer is mounted as in the vehicle
Response	One “Record vitals” action opens a single screen with large fields and number pads; implausible values are flagged before saving
Response measure	In a test with 10 first-time users, at least 8 record the full set within 60 s, with no more than 6 taps per vital sign and no uncorrected error; the System Usability Scale score averages at least 68

QAS-8. Modifiability: replacing the routing provider

Part	Description
Source	Development team
Stimulus	Replace the Google Routes API with another routing provider, for example because the free allowance runs out
Artifact	Routing adapter of the dispatch and routing engine
Environment	Development, after the prototype
Response	The new provider is implemented behind the existing routing interface; nothing outside the adapter changes
Response measure	Only files in the routing adapter and its configuration change, as checked in the pull request; the work takes no more than 2 person-days; all existing routing tests pass against the new adapter

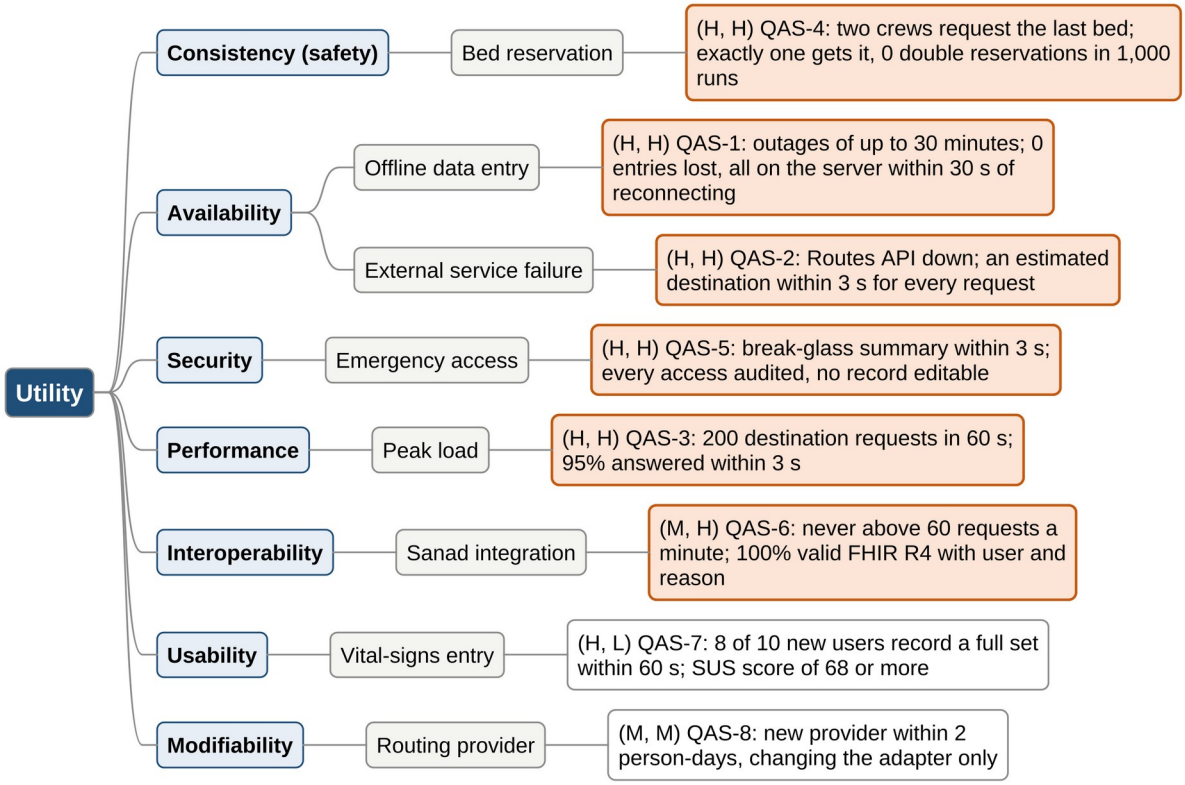
6. Constraints

ID	Type	Constraint
C1	Technical	Three separately deployable parts: the paramedic app on the ambulance computer, the dispatch and routing engine, and the web portal.
C2	Technical	Three external services are fixed: the Google Routes API, Sanad’s HL7 FHIR R4 interface (a mock during the course) and Firebase Cloud Messaging.
C3	Technical	Alerts reach crews only through the ambulance computer; there is no SMS.
C4	Technical	Only synthetic patient data are used in development, tests and demonstrations.
C5	Business	Traffic-aware routing bills as Routes API Pro, with 5,000 free calls a month; a route matrix bills per element, that is, origins times destinations, and holds at most 625 elements.
C6	Business	Free service tiers only, for hosting, database and messaging; Firebase lists Cloud Messaging as no-cost.
C7	Business	Under Egypt’s Personal Data Protection Law (Law No. 151 of 2020), health data are sensitive personal data, and a breach must be reported to the Personal Data Protection Center within 72 hours.
C8	Business	Under the same law, personal data may be transferred, stored or shared outside Egypt only to countries whose protection is no lower than Egypt’s, under a licence or permit from the Center and with the data subject’s consent.
C9	Organisational	RapidMed is to become a module of Sanad, the patient health data management system of the team’s graduation project.
C10	Organisational	Three team members and four milestones: Milestone 1 is due on 23 October 2026, and Milestone 3 delivers a working prototype.
C11	Organisational	All work is submitted through GitHub; documents are PDFs named TeamName_ProjectTitle; diagram sources are kept in the repository; sources are cited in IEEE style.

7. Architecturally significant requirements (ASRs)

The utility tree places each quality attribute scenario under the quality attribute it refines, rated for business value and architectural impact. The table then ranks all the requirements, from the user stories (US), quality attribute scenarios (QAS) and constraints (C), on the same two scales: high (H), medium (M) or low (L). The five rated high for impact are the architecturally significant requirements, ASR-1 to ASR-5.

Utility tree for RapidMed



Legend
Dark blue: the root, overall utility
Light blue: quality attribute
Grey: refinement of the attribute
White: scenario, rated (business value, architectural impact) as H high, M medium or L low
Orange: scenario behind an architecturally significant requirement
Lines run from a node to the nodes that refine it

Rank	Requirement	From	Business value	Architectural impact
ASR-1	Bed reservations are all or nothing, even when requests arrive together or are retried	US-05, QAS-4	H	H
ASR-2	No data lost offline; queued entries sync in order, without duplicates	US-02, QAS-1	H	H
ASR-3	Break-glass access within 3 s, with an append-only audit log that can be reviewed	US-04, US-13, QAS-5, C7	H	H
ASR-4	A destination within 3 s at peak load, and an estimated one when the routing service fails	US-03, QAS-2, QAS-3, C5	H	H
ASR-5	Sanad reached only through FHIR R4, within its rate limit, behind a replaceable adapter	QAS-6, C2, C9	M	H
6	No patient data sent to Google services	C8, US-07	H	M
7	Overrides delivered to the crew and acknowledged	US-11, C3	H	M
8	Free service tiers only	C5, C6	M	M
9	Live map refreshed at least every 5 s	US-10	M	M
10	Routing provider replaceable within 2 person-days	QAS-8	M	M
11	Vital signs recorded in 6 taps or fewer each	US-02, QAS-7	H	L
12	The reasons behind each recommendation shown	US-14	L	L

Why these five will most influence the architecture

1. **ASR-1, atomic bed reservation.** It needs one authoritative record of reservations, updated with conditional writes, and requests that can be retried without effect. It also rules out reserving a bed offline, which puts consistency ahead of availability for that one action: the paramedic app has to separate what it may do offline (record, read cached data) from what must wait for the server (reserve).
2. **ASR-2, offline-first entry.** It needs a persistent store on the ambulance computer, an outbox that replays entries in order, and a server that recognises entries it has already received. Because several crews share the computer, the local store must also be encrypted and separated by user (US-01).
3. **ASR-3, break-glass with audit.** It needs an access path that skips consent but never skips the log, and an audit store kept apart from operational data, to which no application role can make changes. The same log supports the 72-hour breach duty (C7).
4. **ASR-4, a destination under load and failure.** It needs time-outs, a circuit breaker and a cache of travel times around the routing adapter. The free allowance of the Routes API (C5) makes the cache necessary even when the service is up.
5. **ASR-5, integration with Sanad.** It needs an adapter that keeps FHIR at the edge of the engine, with a client-side rate limiter and an episode cache, so that the mock and the real Sanad can be swapped. Its business value is medium because the prototype runs on the mock; its impact is high because it fixes where patient data enter RapidMed.